Manifold Learning for Weather-to-Flood Modeling

UMAP Topological Dimensionality Reduction

MKM Research Labs

Version 1.1 — 13-February-2026

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Contents

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1 Document History

Date	Version	Description	Author
02-April-2025	v1.0	Initial internal beta	David K Kelly
13-February-2026	v1.1	Bank validation protocol; sensitivity analysis	David K Kelly

2 Executive Summary

This document presents the mathematical framework and implementation details for applying Uniform Manifold Approximation and Projection (UMAP) to weather-to-flood modelling. The approach treats high-dimensional atmospheric data as residing on a Riemannian manifold, enabling dimensionality reduction that preserves the essential topological structures governing fluid dynamics over terrain.

The model reduces computational cost by approximately 33× relative to full CFD simulation while maintaining flood prediction RMSE within 0.31 m (compared to 0.24 m for full physics). It achieves 87.3% accuracy in unsupervised storm classification and 42% lower topological distortion than linear methods (PCA).

Key risk: The model is used as an upstream component feeding the hazard curve and PRS pricing models. Errors in dimensionality reduction propagate to flood probability estimates and, consequently, to swap pricing.

3 Model Purpose and Scope

3.1 Purpose

UMAP is applied to reduce the dimensionality of meteorological forecast fields before they enter the hydrodynamic flood model. This enables:

- Real-time processing of HRRR 3 km forecast data within operational latency constraints (<10 minutes)
- Preservation of multi-scale atmospheric structures (synoptic, mesoscale, microscale) that drive flood response
- Uncertainty quantification through ensemble processing in reduced dimensions

3.2 Scope

- **In scope:** Dimensionality reduction of atmospheric fields; mapping to hydrodynamic initial conditions; topology preservation metrics
- **Out of scope:** The hydrodynamic solver itself (shallow water equations); property-level damage estimation; PRS pricing
- **Upstream dependencies:** HRRR forecast data (3 km, hourly)
- **Downstream consumers:** Gauge response model, hazard curve builder, storm intensity distribution

4 Mathematical Framework

4.1 UMAP Core Theory

4.1.1 Local Metric Approximation

For each high-dimensional point x_i in the weather state space, UMAP constructs local neighbourhood relationships. The local connectivity radius is:

$$\rho_i = \min\{d(x_i, x_{ij}) \mid 1 \leq j \leq k,\ d(x_i, x_{ij}) > 0\} \tag{1}$$

The bandwidth parameter σ_i is calibrated to ensure consistent local connectivity:

$$\sum_{j=1}^{k} \exp\left(-\frac{\max(0,\, d(x_i, x_{ij}) - \rho_i)}{\sigma_i}\right) = \log_2(k) \tag{2}$$

This adapts to varying densities in atmospheric data, where specific weather patterns may be densely clustered while others are sparsely distributed.

4.1.2 Fuzzy Topological Representation

UMAP constructs a weighted k-nearest neighbour graph with edge weights representing fuzzy set membership:

$$w_{ij} = \exp\left(-\frac{\max(0,\, d(x_i, x_j) - \rho_i)}{\sigma_i}\right) \tag{3}$$

The directed adjacency matrix A is symmetrised via a probabilistic t-conorm:

$$B = A + A^\top - A \circ A^\top \tag{4}$$

where $\circ$ denotes the Hadamard product. This preserves the topological structure of weather patterns across varying scales.

4.1.3 Cross-Entropy Optimisation

The low-dimensional embedding minimises the cross-entropy between high- and low-dimensional fuzzy simplicial set representations:

$$\mathcal{L} = \sum_{(i,j)\in E} \left[w_h(i,j) \log \frac{w_h(i,j)}{w_l(i,j)} + (1 - w_h(i,j)) \log \frac{1 - w_h(i,j)}{1 - w_l(i,j)} \right] \tag{5}$$

The low-dimensional similarity uses a Student's t-distribution kernel:

$$w_l(i,j) = \left(1 + a \cdot d(y_i, y_j)^{2b}\right)^{-1} \tag{6}$$

with default parameters $a \approx 1.93$, $b \approx 0.79$.

4.2 Multi-Scale Weather Pattern Integration

4.2.1 Scale-Aware Manifold Construction

Weather systems span synoptic to microscale. We construct a hierarchical manifold representation:

$$\mathcal{M}(W(t)) = \bigoplus_{i=1}^{m} \alpha_i\, M_i, \qquad M_i = \mathrm{UMAP}(\mathcal{F}_i(W(t))) \tag{7}$$

where $\mathcal{F}_i$ applies scale-specific filtering and α_i are variance-derived importance weights.

4.2.2 Persistent Homology Filtering

Topological features are filtered via persistent homology:

$$PH(M) = \{(b_i, d_i) \mid i = 1, \ldots, n\} \tag{8}$$

where (b_i, d_i) are birth–death pairs. Features with persistence above threshold τ are retained:

$$M' = \{x \in M \mid \exists (b_i, d_i) \in PH(M) : (d_i - b_i) > \tau,\ x \in \mathrm{Support}(b_i, d_i)\} \tag{9}$$

4.2.3 Weather-to-Flood Mapping

The reduced representation feeds a physics-constrained neural operator:

$$F(t + \Delta t) = \Psi(\mathcal{M}(W(t)), F(t)), \qquad \Psi = \mathcal{D} \circ \mathcal{N} \circ \mathcal{E} \tag{10}$$

where $\mathcal{E}$ encodes the combined state, $\mathcal{N}$ integrates forward in time, and $\mathcal{D}$ decodes subject to physical constraints.

4.3 Uncertainty Quantification

4.3.1 Manifold Distortion Metrics

Local distortion is quantified via trustworthiness $T(k)$ and continuity $C(k)$:

$$T(k) = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^{n} \sum_{j \in U_i^k} (r(i,j) - k) \tag{11}$$

$$C(k) = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^{n} \sum_{j \in V_i^k} (\hat{r}(i,j) - k) \tag{12}$$

4.3.2 Error Propagation

Input uncertainties propagate through the manifold mapping via:

$$\Sigma_{\mathcal{M}} = J_{\mathcal{M}} \, \Sigma_W \, J_{\mathcal{M}}^T \tag{13}$$

where $J_{\mathcal{M}}$ is the Jacobian of the UMAP embedding and Σ_W is the weather state covariance matrix.

4.3.3 Ensemble Approach

Probabilistic flood prediction uses N_e ensemble members:

$$P(F(t + \Delta t) \in A) = \frac{1}{N_e} \sum_{i=1}^{N_e} \mathbf{1}_A(F^{(i)}(t + \Delta t)) \tag{14}$$

5 Input Parameters and Calibration

5.1 UMAP Hyperparameters

Parameter	Role	Range	Default
n_neighbors	Local vs global structure balance	15–50	30
min_dist	Spacing between similar weather states	0.1–0.5	0.25
metric	Distance function	Euclidean, cosine	Euclidean
n_components	Target dimensionality	3–10	5

Table 1: UMAP hyperparameters and calibration ranges

5.2 Weather Input Fields

Variable	Source	Resolution
Surface pressure	HRRR forecast	3 km, hourly
Temperature (2 m)	HRRR forecast	3 km, hourly
Specific humidity	HRRR forecast	3 km, hourly
Wind vectors (10 m)	HRRR forecast	3 km, hourly
Terrain elevation	Static DEM	250 m
Slope and aspect	Derived	250 m
Soil properties	Static maps	1 km

Table 2: Input data specification

5.3 Calibration Protocol

1. Fit UMAP on 3 years of historical HRRR data (2022–2024)
2. Optimise $n\_neighbors$ and min_dist via grid search over trustworthiness $T(k) > 0.95$ and continuity $C(k) > 0.90$
3. Validate storm classification accuracy on held-out labelled set (500 events)
4. Cross-validate flood prediction RMSE against full CFD baseline

6 Implementation Details

6.1 Computational Workflow

1. Ingest raw HRRR forecast data at 3 km resolution
2. Normalise and bias-correct against historical observation–forecast pairs
3. Apply quality control (missing value detection, anomaly flagging)
4. Apply scale-aware UMAP dimensionality reduction
5. Map reduced representations to hydrodynamic initial conditions
6. Execute ensemble flood simulations (shallow water equations)
7. Aggregate into probabilistic flood predictions
8. Project impacts onto property vulnerability assessments

6.2 Hardware Acceleration

- Nearest-neighbour search parallelised via FAISS (GPU)
- Stochastic gradient descent for embedding optimisation (GPU)
- Batched ensemble processing across multiple GPUs
- Approximate 40× speedup versus CPU implementation

6.3 Operational Parameters

Parameter	**Value**
Update frequency	Every 6 hours (aligned with HRRR)
Forecast horizon	48 hours, hourly resolution
Spatial resolution	250 m (downscaled from 3 km)
Ensemble size	50 members
Processing latency	<10 minutes end-to-end

Table 3: Operational deployment parameters

7 Sensitivity Analysis

The sensitivity of model outputs to key hyperparameters was assessed by sweeping each parameter across its calibration range while holding others at defaults.

7.1 *n_neighbors* Sensitivity

n_neighbors	Trustworthiness	Continuity	Storm Accuracy	Flood RMSE (m)
15	0.972	0.891	84.1%	0.35
20	0.968	0.912	85.8%	0.33
30	0.961	0.934	87.3%	0.31
40	0.953	0.941	86.9%	0.32
50	0.944	0.948	85.2%	0.33

Table 4: Sensitivity to *n_neighbors* (default = 30)

Finding: Flood RMSE is minimised at *n_neighbors* = 30. Performance degrades modestly (± 0.04 m) across the full range. Trustworthiness decreases with larger neighbourhoods (over-smoothing of local structure), while continuity increases (better global preservation).

7.2 *min_dist* Sensitivity

min_dist	Trustworthiness	Continuity	Storm Accuracy	Flood RMSE (m)
0.05	0.974	0.918	88.1%	0.33
0.10	0.969	0.926	87.8%	0.32
0.25	0.961	0.934	87.3%	0.31
0.40	0.955	0.939	86.4%	0.32
0.50	0.949	0.942	85.7%	0.34

Table 5: Sensitivity to *min_dist* (default = 0.25)

Finding: Model is relatively insensitive to *min_dist* within the 0.10–0.40 range (RMSE varies ± 0.02 m). Very small values (<0.05) risk overfitting to noise; large values (>0.50) lose discriminative power between distinct storm types.

7.3 *n_components* Sensitivity

n_components	Trustworthiness	Continuity	Storm Accuracy	Flood RMSE (m)
3	0.942	0.907	82.6%	0.38
5	0.961	0.934	87.3%	0.31
7	0.969	0.941	87.8%	0.30
10	0.974	0.946	87.5%	0.31

Table 6: Sensitivity to target dimensionality (default = 5)

Finding: The most material parameter. Reducing from 5 to 3 dimensions increases RMSE by 23% (0.31 to 0.38 m), indicating important flood-relevant structure is lost. Increasing beyond 5 provides marginal improvement. The intrinsic dimension of the weather-to-flood manifold is approximately 5.

7.4 Ensemble Size Sensitivity

N_e	90% CI Width (m)	CRPS	Latency (min)
10	0.82	0.187	2.1
20	0.71	0.164	3.8
50	0.63	0.148	8.7
100	0.61	0.145	17.2

Table 7: Sensitivity to ensemble size (default = 50)

Finding: Diminishing returns beyond $N_e = 50$. The 90% confidence interval width decreases by only 3% when doubling from 50 to 100, while latency doubles. The default of 50 provides adequate uncertainty quantification within the 10-minute operational window.

8 Validation and Backtesting

8.1 Weather Pattern Classification

UMAP's ability to identify coherent meteorological patterns was evaluated on a labelled dataset of 500 historical storm events:

Method	**Classification Accuracy**	**Silhouette Score**
PCA	71.8%	0.34
t-SNE	79.4%	0.48
UMAP	87.3%	0.61

Table 8: Unsupervised storm classification performance

8.2 Flood Prediction Accuracy

Method	**RMSE (m)**	**Timing Error (h)**	**Relative Cost**
Full CFD	0.24	1.2	100×
PCA + Physics	0.43	2.8	2.5×
UMAP + Physics	0.31	1.7	3.0×

Table 9: Flood prediction performance comparison

8.3 Topological Preservation

Topology preservation was quantified using the Wasserstein distance between persistence diagrams:

$$W_p(PD_{\text{orig}}, PD_{\text{red}}) = \left(\inf_{\gamma} \sum_{x \in PD_{\text{orig}}} \|x - \gamma(x)\|_{\infty}^{p} \right)^{1/p} \tag{15}$$

UMAP achieved a 42% reduction in topological distortion compared to PCA, confirming superior preservation of watershed boundaries and frontal structures.

8.4 Historical Backtesting

The model was backtested against 12 major flood events (2020–2024) in the Thames catchment. Results are summarised as:

- 11/12 events: peak water level predicted within ± 0.4 m
- 10/12 events: timing of peak within ± 3 hours
- 1 event (Feb 2022): underpredicted by 0.7 m due to unprecedented antecedent soil saturation not captured in the training data

9 Model Limitations and Known Weaknesses

1. **Stationarity assumption:** UMAP is trained on historical data and assumes the statistical relationship between weather patterns and flood response is stationary. Climate change may invalidate this over multi-decade horizons.
2. **Static embedding:** Standard UMAP does not model temporal evolution. Sequential embeddings are used as a workaround but may miss rapid regime transitions (e.g., sudden convective initiation).
3. **Conservation law compliance:** The dimensionality reduction does not explicitly enforce mass/energy/momentum conservation. Post-processing validation is applied but is not guaranteed to detect all violations.
4. **Training data coverage:** Extreme events beyond the historical training set (2022–2024) may produce embeddings in under-represented regions of the manifold, leading to higher prediction uncertainty.
5. **Ensemble calibration:** The ensemble spread may underestimate true uncertainty for events near the boundary of the training distribution.
6. **Spatial resolution gap:** Downscaling from 3 km atmospheric to 250 m flood resolution introduces interpolation uncertainty not fully captured in the error budget.

10 Change History

Date	Version	Change Description
02-Apr-2025	1.0	Initial document: mathematical framework, implementation architecture, experimental validation

Date	**Version**	**Change Description**
13-Feb-2026	1.1	Added bank validation protocol sections: executive summary, sensitivity analysis (Tables 4–7), backtesting results, model limitations, input calibration specification

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